

Ivan Pavlov

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Education

Moscow Institute of Physics and Technologies

Aug 2022 – present

BSc in Applied Mathematics and Physics, Landau School of Physics and Research

Educational Program "Quantum Field Theory, String Theory and Mathematical Physics"

GPA: 3,44

Thesis advisor: Alexander Gorsky

Research topic: DSSYK model and duality with sin-dilaton gravity

Selected coursework

Mathematics Differential Geometry, Topology, Lie Groups,
Lie Algebras and Representations, Classical
Integrable Systems, Integrable Lattice Models

Physics Classical Mechanics, Classical Field Theory,
Quantum Mechanics, QFT, CFT, General
Relativity, Cosmology, Statistical Mechanics

Research Experience

Workshop on Theoretical Physics organized by Leonid Levitov (MIT)

2024, 2025

Project: "Soundwaves in Topological Insulators with Non-zero Chern Number"

Aug 2024

Advisor: Ilya Esterlis

- [Slides](#) of the final presentation

Project: "Superconductors with non-unitary triplet pairing in external magnetic field"

Aug 2025

Advisors: Vlad Kozii, Maxim Dzero

- [Slides](#) of the final presentation

Academic Projects, Schools and Talks

Project "Mentor"

Sep - Dec 2023

- Topic: QCD beta-function ([Slides](#) of the final presentation)

ITMP Winter School

Feb 2025

Section: Supersymmetric Quantum Mechanics

Advisor: Dmitry Bykov (MIAS, ITMP)

- Talk on superspace formalism, ([notes](#) in Russian)
- Talk on SUSY sigma-models, ([notes](#) in Russian)

Spring School on Mathematical Physics in SkolTech

Apr 2025

Section: 6-vertex Model

Advisor: Nikita Ignatyuk (SkolTech)

- Talk on the thermodynamic limit of Bethe Ansatz equations ([notes](#) in Russian, [video](#) in Russian)

Summer School and Workshop "Vertex Algebras & Integrable Systems"

July 2025

Actively participated in exercise sessions; completed problem sets on Cluster Algebras

- My [solutions](#) to the exercises from a course on Cluster Algebras

Topics in Classical Field Theory (part of a course by Alexander Gorsky)

Apr 2025

- Talk on the geometry of Yang-Mills theory and ADHM construction ([notes](#) and [video](#) in Russian)

Other achievements

ITMP Universiade 2025 - Awardee

Apr 2025

- Qualified in a problem-solving contest, gave a short talk on $SU(2)$ representations in the final contest

School olympiads - Multiple Wins and Awards

2021-2022

- "Phystech" - Winner (physics 2021, mathematics 2022)
- Other national olympiads - 9 prizes in 2022

Extracurricular & Leadership

Volunteering, Event Staff	2022-2025
– Played a key role in constructing the campus’s first rehearsal studio and logistics for large-scale concerts (“Physicist’s Day”, “Phystech’s Birthday”) as a staff member for mipt.live.	
MIPT Musical community	2022-2025
– Founder and drummer of bands “Dva Kamnya” and “The Cusp”. Participated in every major concert at MIPT	
Fluent in English	